Database Used: Sakila (MySQL Workbench)
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Assignment: SQL_Assignment

Q1. Display the first name, last name, and rental date of all customers who rented a film.

```
USE sakila;
SELECT c.first_name, c.last_name, r.rental_date
FROM customer c
JOIN rental r ON c.customer_id = r.customer_id;
```

Q2. Display the first name, last name, and address of all customers whose district is either Alberta or Texas.

```
SELECT c.first_name, c.last_name, a.address
FROM customer c
JOIN address a ON c.address_id = a.address_id
WHERE a.district IN ('Alberta', 'Texas');
```

Q3. List all films along with their category names, including films that may not belong to any category. Sort by category name.

```
SELECT f.title, c.name AS category_name
FROM film f
LEFT JOIN film_category fc ON f.film_id = fc.film_id
LEFT JOIN category c ON fc.category_id = c.category_id
ORDER BY category_name;
```

Q4. Display the number of rentals per store.

```
SELECT s.store_id, COUNT(r.rental_id) AS total_rentals
FROM rental r
JOIN inventory i ON r.inventory_id = i.inventory_id
JOIN store s ON i.store_id = s.store_id
GROUP BY s.store_id;
```

Q5. List the names of categories that have more than 50 films.

```
SELECT c.name, COUNT(fc.film_id) AS total_films
FROM category c
JOIN film_category fc ON c.category_id = fc.category_id
GROUP BY c.name
HAVING COUNT(fc.film_id) > 50;
```

Q6. Display the film titles rented by the customer named 'Mary Smith'.

```
SELECT title
FROM film
WHERE film_id IN (
   SELECT i.film_id
   FROM inventory i
   JOIN rental r ON i.inventory_id = r.inventory_id
   WHERE r.customer_id = (
      SELECT customer_id
      FROM customer
      WHERE first_name = 'Mary' AND last_name = 'Smith'
   )
);
```

Q7. Display the total rental income generated by each film.

```
SELECT f.title, SUM(p.amount) AS total_income
FROM film f
JOIN inventory i ON f.film_id = i.film_id
JOIN rental r ON i.inventory_id = r.inventory_id
JOIN payment p ON r.rental_id = p.rental_id
GROUP BY f.title;
```

Q8. Display the total number of films each actor has acted in.

```
SELECT a.first_name, a.last_name, COUNT(fa.film_id) AS total_films
FROM actor a
JOIN film_actor fa ON a.actor_id = fa.actor_id
GROUP BY a.actor_id;
```

Q9. List all rentals with a case statement that displays 'New Customer' if customer_id is greater than 300, otherwise 'Regular Customer'.

```
SELECT rental_id, customer_id,
CASE
   WHEN customer_id > 300 THEN 'New Customer'
   ELSE 'Regular Customer'
END AS customer_type
FROM rental;
```

Q10. Find all films with titles that start with the letter 'A'. Display the film title and release year.

```
SELECT DISTINCT f.title, f.release_year
FROM film f
JOIN inventory i ON f.film_id = i.film_id
WHERE f.title LIKE 'A%';
```

Q11. Create a New Database and Perform Given Operations

```
CREATE DATABASE student_db;
USE student_db;
CREATE TABLE Student (
   Student_ID INT,
   Marks INT);
INSERT INTO Student VALUES (1, 85);
INSERT INTO Student VALUES (2, 90);
ALTER TABLE Student
RENAME COLUMN Marks TO Score;
DROP TABLE Student;
DROP DATABASE student_db;
```